

QUESTION 1:

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main(){
    int camp=3;
    int dep=3;
    int **std_num=new int*[camp];
    for(int i=0;i<camp;i++){
        | *(std_num+i)=new int[dep];
    }
    for(int i=0;i<camp;i++)
    {
        | if(i==0)
        |
        | { //first camp
        |   for(int j=0;j<dep;j++)
        |   {
        |
        |   if(j==0)
        |   {
        |
        |       cout<<"enter students for Peshawar Campus - COMP SCI ";
        |       cin>>*(*(std_num+i)+j);
        |   }
    }
```

```

    }
    if(j==1)
    {
        cout<<"enter students for Peshawar Campus - SOFT ENGR ";
        cin>>*(*(std_num+i)+j);
    }
    if(j==2)
    {
        cout<<"enter students for Peshawar Campus - DATA SCI ";
        cin>>*(*(std_num+i)+j);
    }
    }
}

```

```

if(i==1)
{
    //second camp
    for(int j=0;j<dep;j++)
    {
        if(j==0)
        {
            cout<<"enter students for Islamabad Campus - COMP SCI ";
            cin>>*(*(std_num+i)+j);
        }
        if(j==1)
        {
            cout<<"enter students for Islamabad Campus - SOFT ENGR ";
            cin>>*(*(std_num+i)+j);
        }
        if(j==2)
        {
            cout<<"enter students for Islamabad Campus - DATA SCI ";
            cin>>*(*(std_num+i)+j);
        }
    }
}
}

```

```
if(i==2)
{    //third camp
    for(int j=0;j<dep;j++)
    {
        if(j==0)
        {
            cout<<"enter students for Lahore Campus - COMP SCI ";
            cin>>*(*(std_num+i)+j);
        }
        if(j==1)
        {
            cout<<"enter students for Lahore Campus - SOFT ENGR ";
            cin>>*(*(std_num+i)+j);
        }
        if(j==2)
        {
            cout<<"enter students for Lahore Campus - DATA SCI ";
            cin>>*(*(std_num+i)+j);
        }
    }
}
```

```
cout<<"-----"<<endl;
cout<<"Campus/Department"<<"\t"<<"CS  SE  DS"<<endl;
cout<<"-----"<<endl;
for(int i=0;i<camp;i++){
    if(i==0){
        cout<<"Peshawar Campus"<<"\t\t";

        for(int j=0;j<dep;j++){
            cout<<*(*std_num+i+j)<<" ";
        }
        cout<<endl;
    }
    else if(i==1){
        cout<<"Islamabad Campus"<<"\t";

        for(int j=0;j<dep;j++){
            cout<<*(*std_num+i+j)<<" ";
        }
        cout<<endl;
    }
    else if(i==2){
        cout<<"Lahore Campus"<<"\t\t";
```

```

        else if(i==2){
            cout<<"Lahore Campus"<<"\t\t";

            for(int j=0;j<dep;j++){
                cout<<*(*(std_num+i)+j)<<" ";
            }
            cout<<endl;
        }
    }

    for (int i=0;i<camp;i++){
        delete[] std_num[i];
    }
    delete[] std_num;
    return 0;
}

```

OUTPUT:

```

enter students for Peshawar Campus - COMP SCI 37
enter students for Peshawar Campus - SOFT ENGR 67
enter students for Peshawar Campus - DATA SCI 65
enter students for Islamabad Campus - COMP SCI 38
enter students for Islamabad Campus - SOFT ENGR 90
enter students for Islamabad Campus - DATA SCI 76
enter students for Lahore Campus - COMP SCI 34
enter students for Lahore Campus - SOFT ENGR 56
enter students for Lahore Campus - DATA SCI 75
-----
Campus/Department      CS   SE   DS
-----
Peshawar Campus        37   67   65
Islamabad Campus       38   90   76
Lahore Campus          34   56   75

```

QUESTION 2:

SOURCE CODE:

```
#include <iostream>
using namespace std;
int main(){
    int row=3;
    int col=3;
    bool **enemy=new bool*[row];
    for(int i=0;i<row;i++){
        *(enemy+i)=new bool[col];
    }

    for(int i=0;i<row;i++){
        for(int j=0;j<col;j++){
            cin>>*(*(enemy+i)+j);
        }
    }
    int enemy_count=0;
    for(int i=0;i<row;i++){
        for(int j=0;j<col;j++){
            if(*(*(enemy+i)+j)==1){
                enemy_count++;
            }
        }
    }
    cout<<"Total live enemies: "<<enemy_count<<endl;
```

```

int i;
int j;
int hit_count=0;
do{
    cout<<"Enter row and column "<<endl;
    cin>>i;
    cin>>j;
    if(*(enemy+i)+j)==1){
        *(enemy+i)+j=0;
        cout<<"hit. enemy destroyed"<<endl;
        enemy_count--;
        hit_count++;
    }
}while(enemy_count!=0);
cout<<"All enemies are destroyed"<<endl;
cout<<"total hits required "<<hit_count<<endl;

for(int p=0;p<row;p++){
    for(int q=0;q<col;q++){
        cout<<*(enemy+i)+j<<" ";
    }
    cout<<endl;
}

```

```

for(int k=0;k<row;k++){
    delete[] enemy[i];
}
delete[] enemy;
}

```

OUTPUT:

```
1
0
1
1
1
0
0
1
1
Total live enemies: 6
Enter row and column
0
0
hit. enemy destroyed
Enter row and column
0
2
hit. enemy destroyed
Enter row and column
1
0
hit. enemy destroyed
Enter row and column
1
1
hit. enemy destroyed
```



```
Enter row and column
1
2
Enter row and column
2
1
hit. enemy destroyed
Enter row and column
2
2
hit. enemy destroyed
All enemies are destroyed
total hits required 6
0 0 0
0 0 0
0 0 0
```